Unity (C#) - Manual Play Test Plan

Semester 1 Capstone - Code Complete Checklist

VIDEO GAME CREATION

This is a manual, play-by-hand test plan - no automated tests, no unit tests, just running the actual game and confirming each behavior. Work through every section in order; a section near the bottom assumes everything above it already passes. If anything fails, fix it, rebuild, and re-run that section (and anything after it that could plausibly be affected) before continuing - don't skip ahead on a failing section.

When every box below is checked Pass, this project is Code Complete and becomes the master model for the next engine. Any Fail should be logged (what you did, what you expected, what actually happened) before moving on to fix it - that log is worth keeping even after the bug's fixed.

0. Prerequisites

☐ **All Pass** ☐ **Had Failures**

Confirm these before testing anything else - a failure here will cause confusing failures everywhere below, for reasons that have nothing to do with the actual feature being tested.

#	Steps	Expected Result	Result
1	Open the project in the Unity Editor and let it fully compile.	No compiler errors. No compiler warnings either (we specifically cleaned up the FindObjectsByType obsolete-API warnings - if new warnings appear, note them before continuing).	☐ Pass ☐ Fail
2	Check every GameObject and prefab against the Setup Guide's field tables (GameManager, EnemySpawner, Enemy, Bullet, Player, PlayerMovement, HUD, Shield x however many you placed, PowerUp, PowerUpSpawner, FormationController).	Every Inspector field listed in the Setup Guide is assigned - nothing is left on "None (...)".	☐ Pass ☐ Fail
3	Confirm the Rigidbody2D / Collider2D pairings from the Setup Guide specifically: Bullet has Rigidbody2D, Shield has Rigidbody2D (Kinematic), PowerUp has Rigidbody2D (Kinematic), Player has a Collider2D (Is Trigger).	All four are present - these are the ones that silently do nothing if missed, rather than throwing an error.	☐ Pass ☐ Fail

#	Steps	Expected Result	Result
4	Confirm both of Player's AudioSource components have Play On Awake UNCHECKED.	Nothing plays automatically before you press anything.	☐ Pass ☐ Fail

Tester Comments

1. Title Screen & Start

☐ **All Pass** ☐ **Had Failures**

Covers Lesson 14's title-screen stretch feature and this session's Space-key fire-guard fix.

#	Steps	Expected Result	Result
5	Press Play. Don't touch anything yet.	"Kestrel Watch / Press SPACE to begin your watch" is shown, centered, no enemies on screen, no sound playing.	☐ Pass ☐ Fail
6	Press Space to start.	The title text disappears, the game-start sound plays once, Wave 1's enemies spawn (5 of them), and the player does NOT also fire a bullet on that same press.	☐ Pass ☐ Fail
7	After starting, press R.	Nothing happens - R only works once the game has ended (see Section 6).	☐ Pass ☐ Fail

Tester Comments

2. Core Movement & Shooting (Level 1/2)

☐ **All Pass** ☐ **Had Failures**

The original Lesson 2-3 mechanics, unchanged in behavior.

#	Steps	Expected Result	Result
8	Move with the arrow keys in all four directions, then release.	Player moves smoothly, clamped to the play area; sprite changes per direction while moving, returns to the default sprite when idle; a walking beep loops while moving and stops when idle; the sprite blinks on/off while moving.	☐ Pass ☐ Fail

#	Steps	Expected Result	Result
9	Hold Space to fire repeatedly.	A bullet spawns above the player on a short cooldown (not one per frame), travels upward, and is destroyed off the top of the screen if it misses. Each shot plays a sound and shows a brief yellow flash/scale-up on the bullet that fades after about a tenth of a second.	☐ Pass ☐ Fail
10	Fire at an enemy and hit it.	The enemy flickers (flash/normal/flash), an explosion appears at its position and grows slightly before disappearing, a hit sound plays, and the enemy and the bullet are both gone shortly after.	☐ Pass ☐ Fail

Tester Comments

3. Waves, Score, Lives (Level 2)

☐ **All Pass** ☐ **Had Failures**

Confirms the wave/score/lives systems and the historical Stalemate-bug fix all still hold under normal play.

#	Steps	Expected Result	Result
11	Destroy every enemy in Wave 1 without letting any escape.	A "Wave 2!" message flashes briefly, a wave-clear sound plays, and Wave 2's enemies (6 of them) spawn - visibly faster than Wave 1's.	☐ Pass ☐ Fail
12	Let an enemy reach the bottom of the screen instead of shooting it.	A life is lost, a life-lost sound plays, and that enemy still counts toward clearing the wave (this is the Lesson 10/11 Stalemate fix - a wave must be able to end via escapes, not only via kills).	☐ Pass ☐ Fail
13	Clear a wave using a mix of kills AND escapes in the same wave.	The wave still advances correctly once every enemy is accounted for, whichever way each one left the screen.	☐ Pass ☐ Fail
14	Check the HUD (score/lives/wave) throughout.	All three update immediately and correctly after every kill, escape, and wave change - no lag, no stale numbers.	☐ Pass ☐ Fail
15	Lose all 3 lives.	"Game Over - Press R to Restart" fades in (not an instant pop) over about half a second, centered on screen, and a game-over sound plays once.	☐ Pass ☐ Fail
16	Clear all 3 waves without losing.	"You Win! Press R to Restart" fades in the same way, and a win sound plays once.	☐ Pass ☐ Fail

Tester Comments

4. Restart

☐ All Pass ☐ Had Failures

Confirms ResetGame() actually returns the game to a true starting state, including the newer Shield-reset fix.

#	Steps	Expected Result	Result
17	After a win or loss, press R.	Score resets to 0, lives to 3, wave to 1, and Wave 1's 5 enemies spawn fresh - not wherever the previous run left off.	☐ Pass ☐ Fail
18	Restart after at least one Shield has taken damage or been fully broken in the previous run.	Every Shield is back to its starting hit count and original color, and any Shield that had broken (deactivated) is visible and active again.	☐ Pass ☐ Fail

Tester Comments

5. Level 3 Polish - Variety & Feel

☐ All Pass ☐ Had Failures

The "pick any that interest you" options from Lessons 4-10 that the Complete Codebase builds in as a full set rather than a student's personal pick of one or two.

#	Steps	Expected Result	Result
19	Watch a full wave of normal (non-bonus) enemies spawn.	Enemies visibly alternate between two shades of red/pink in a striped pattern by spawn order, not all one uniform color.	☐ Pass ☐ Fail
20	Destroy several normal enemies in a row and watch the score go up each time.	Each kill awards a different point value somewhere in the 5-15 range, not always the same number.	☐ Pass ☐ Fail
21	Watch several enemies descend in lane mode (an odd-numbered wave) over a few seconds.	Enemies visibly move at slightly different speeds from each other, not in perfect lockstep.	☐ Pass ☐ Fail
22	Watch the title screen and the win/lose text specifically for alignment.	All of GameManager's OnGUI text (title, win, lose, wave-announcement) is horizontally centered in its box, not left-stuck to the edge.	☐ Pass ☐ Fail

Tester Comments

6. Stretch Feature - Bonus Enemy

☐ All Pass ☐ Had Failures

Lesson 14 stretch option 1 of 5.

#	Steps	Expected Result	Result
23	Reach any even-numbered wave (Wave 2) and look at the last enemy to spawn.	Exactly one enemy in the wave is visibly gold-tinted, overriding the striping pattern from Section 5.	☐ Pass ☐ Fail
24	Destroy that gold enemy specifically and note the score change.	It awards 50 points - clearly more than a normal kill - and its speed was visibly a bit faster than its row-mates.	☐ Pass ☐ Fail
25	Check an odd-numbered wave (Wave 1 or 3).	No bonus enemy appears - only even waves get one.	☐ Pass ☐ Fail

Tester Comments

7. Stretch Feature - Formation Marching

☐ All Pass ☐ Had Failures

Lesson 14 stretch option 2 of 5 - also where this session's xBoundary bug and the Enemy-speed-variance interaction were found.

#	Steps	Expected Result	Result
26	Reach Wave 2 and watch the enemies' overall movement pattern (not individual speed variance from Section 5).	The whole wave moves together side to side as one group, rather than each enemy independently descending down its own lane.	☐ Pass ☐ Fail
27	Watch what happens when the formation reaches the edge of its lane.	The entire formation steps down together and reverses direction - it does NOT immediately plummet to the bottom of the screen on the very first frame.	☐ Pass ☐ Fail
28	Compare Wave 2's movement feel to Wave 1 and Wave 3.	Wave 2 clearly reads as a different, more organized threat - waves 1 and 3 are lane descent, wave 2 is formation marching.	☐ Pass ☐ Fail

Tester Comments

8. Stretch Feature - Destructible Shields

☐ All Pass ☐ Had Failures

Lesson 14 stretch option 3 of 5 - also where this session's Rigidbody2D and double-kill-count bugs were found.

#	Steps	Expected Result	Result
29	Let an enemy descend into a Shield.	The enemy is destroyed, the Shield's color visibly changes (first damage color), and the Shield's remaining-hit count goes down by exactly one.	☐ Pass ☐ Fail
30	Hit the same Shield enough times to use it up.	The Shield changes color again on the second hit, then disappears entirely once its hit count reaches zero.	☐ Pass ☐ Fail
31	Check that a Shield kill still counts toward the wave.	enemiesRemaining still reaches zero and the wave still clears normally when a Shield (not a Bullet) is what destroyed the last enemy - this is the stalemate-twin bug fix.	☐ Pass ☐ Fail
32	If possible, arrange for a Bullet and a Shield to reach the same enemy at nearly the same instant (fire just as an enemy reaches a Shield).	Only ONE kill is registered for that enemy, not two - the wave does not clear early with another enemy still visibly on screen.	☐ Pass ☐ Fail

Tester Comments

9. Stretch Feature - Power-Up

☐ All Pass ☐ Had Failures

Lesson 14 stretch option 4 of 5 - also where this session's Rigidbody2D and Player-Collider2D bugs were found.

#	Steps	Expected Result	Result
33	Wait for a power-up to drop (every 8-15 seconds while the game is active) and fly the Player into it.	The power-up disappears on contact and the Player's fire rate is noticeably faster for a few seconds.	☐ Pass ☐ Fail
34	Let the rapid-fire window expire on its own.	Fire rate returns to normal afterward without you having to do anything.	☐ Pass ☐ Fail

#	Steps	Expected Result	Result
35	Let a power-up fall all the way to the bottom of the screen without collecting it.	It disappears on its own rather than lingering forever.	☐ Pass ☐ Fail

Tester Comments

10. Title/Game-Over Screen

☐ **All Pass** ☐ **Had Failures**

Lesson 14 stretch option 5 of 5 - covered functionally in Sections 1 and 3; this section is specifically about the fade/centering polish from Section 5 applying here too.

#	Steps	Expected Result	Result
36	Re-check the title screen, the win screen, and the lose screen back to back in one sitting.	All three use the same centered, faded-in presentation - none of them pops in instantly or sits off-center while the others are polished.	☐ Pass ☐ Fail

Tester Comments

11. Full Playthrough

☐ **All Pass** ☐ **Had Failures**

One clean run start to finish, paying attention to the whole experience rather than one feature at a time.

#	Steps	Expected Result	Result
37	Play one complete run: title screen, all 3 waves (mixing kills, escapes, and at least one Shield and one power-up encounter), and a win or loss at the end, followed by a restart.	Nothing about the experience feels broken, silent when it shouldn't be, or visually wrong - this is the final sanity check before calling the project Code Complete.	☐ Pass ☐ Fail

Tester Comments

Sign-Off

☐ **Code Complete** ☐ **Not Yet**

Date tested: _____ Tested by:_____